Unit 2 — Solving Logarithmic Inequalities — Worked Solutions

10 questions, Nov 3, 2026. For the teacher — do not give to the student before the retake.
Method in every question: **domain first**, then drop the log (base > 1, so the direction stays), then intersect with the domain. "log" with no base is base 10.

Q1 — $\log(x + 2) > \log(3x - 4)$

Domain: $x + 2 > 0$ gives $x > -2$; $3x - 4 > 0$ gives $x > 4/3$. So the domain is $x > 4/3$.

Drop the log: $x + 2 > 3x - 4 \rightarrow 6 > 2x \rightarrow x < 3$.

Intersect: $x > 4/3$ and $x < 3$.

> **Answer: $(4/3, 3)$**

Q2 — $\ln(x^2 + 4) > \ln(5x - 6)$

Domain: $x^2 + 4 > 0$ for every x; $5x - 6 > 0$ gives $x > 6/5$. So the domain is $x > 6/5$.

Drop the log: $x^2 + 4 > 5x - 6 \rightarrow x^2 - 5x + 10 > 0$.

Check the quadratic: discriminant $25 - 4(1)(10) = -15 < 0$, and the leading coefficient is positive, so $x^2 - 5x + 10 > 0$ for all x. It never flips.

Intersect: the domain alone decides it.

> **Answer: $(6/5, \infty)$**

Q3 — $\log(x^2 - 5x + 6) < \log(x + 2)$

Domain: $x^2 - 5x + 6 = (x - 2)(x - 3) > 0$ gives $x < 2$ or $x > 3$; $x + 2 > 0$ gives $x > -2$. So the domain is $(-2, 2) \cup (3, \infty)$.

Drop the log: $x^2 - 5x + 6 < x + 2 \rightarrow x^2 - 6x + 4 < 0$.

Roots: $x = (6 \pm \sqrt{(36 - 16)}) / 2 = 3 \pm \sqrt{5}$. A upward parabola is below zero between its roots: $(3 - \sqrt{5}, 3 + \sqrt{5})$.

Intersect: cut $(3 - \sqrt{5}, 3 + \sqrt{5})$ by the domain $(-2, 2) \cup (3, \infty)$.

> **Answer: $(3 - \sqrt{5}, 2) \cup (3, 3 + \sqrt{5})$**

Q4 — $f(x) = \log(x^2 - 7x + 10)$, $f(x) > 0$

Domain: $x^2 - 7x + 10 = (x - 2)(x - 5) > 0$ gives $x < 2$ or $x > 5$.

Drop the log: $\log(\ldots) > 0 = \log 1$, so $x^2 - 7x + 10 > 1 \rightarrow x^2 - 7x + 9 > 0$.

Roots: $x = (7 \pm \sqrt{(49 - 36)}) / 2 = (7 \pm \sqrt{13}) / 2$. An upward parabola is above zero outside its roots: $x < (7 - \sqrt{13})/2$ or $x > (7 + \sqrt{13})/2$.

Intersect: $(7 - \sqrt{13})/2 \approx 1.70 < 2$ and $(7 + \sqrt{13})/2 \approx 5.30 > 5$, so the domain cuts nothing off.

> **Answer: $(-\infty, (7 - \sqrt{13})/2) \cup ((7 + \sqrt{13})/2, \infty)$**

Q5 — $g(x) = \log(x^2 - 2x - 8) - \log(x + 2)$, $g(x) > 0$

Domain: $x^2 - 2x - 8 = (x - 4)(x + 2) > 0$ gives $x < -2$ or $x > 4$; $x + 2 > 0$ gives $x > -2$. So the domain is $x > 4$.

Combine the logs: $\log((x^2 - 2x - 8) / (x + 2)) > 0$. Since $x^2 - 2x - 8 = (x - 4)(x + 2)$, the fraction simplifies to $x - 4$.

Drop the log: $x - 4 > 1 \rightarrow x > 5$.

Intersect: $x > 5$ is already inside $x > 4$.

Answer: (5, ∞)

Q6 — $h(x) = \ln(x^2 - 4) - \ln(x - 1)$, $h(x) < 0$

Domain: $x^2 - 4 > 0$ gives $x < -2$ or $x > 2$; $x - 1 > 0$ gives $x > 1$. So the domain is $x > 2$.

Combine the logs: $\ln((x^2 - 4) / (x - 1)) < 0$, so $(x^2 - 4) / (x - 1) < 1$.

Solve: on the domain $x - 1 > 0$, multiply through: $x^2 - 4 < x - 1 \rightarrow x^2 - x - 3 < 0 \rightarrow$ roots $(1 \pm \sqrt{13})/2$, so $(1 - \sqrt{13})/2 < x < (1 + \sqrt{13})/2$.

Intersect: with $x > 2$.

Answer: $(2, (1 + \sqrt{13})/2)$

Q7 — $f(x) = \ln(x^2 - 3x - 4) - \ln(x + 2)$, $f(x) > 0$

Domain: $x^2 - 3x - 4 = (x - 4)(x + 1) > 0$ gives $x < -1$ or $x > 4$; $x + 2 > 0$ gives $x > -2$. So the domain is $(-2, -1) \cup (4, \infty)$.

Combine the logs: $\ln((x^2 - 3x - 4) / (x + 2)) > 0$, so $(x^2 - 3x - 4) / (x + 2) > 1$.

Solve: $(x^2 - 3x - 4) / (x + 2) - 1 > 0 \rightarrow (x^2 - 4x - 6) / (x + 2) > 0$. Critical points -2, $2 - \sqrt{10} \approx -1.16$, $2 + \sqrt{10} \approx 5.16$. Sign chart gives $(-2, 2 - \sqrt{10}) \cup (2 + \sqrt{10}, \infty)$.

Intersect: with the domain $(-2, -1) \cup (4, \infty)$.

Answer: $(-2, 2 - \sqrt{10}) \cup (2 + \sqrt{10}, \infty)$

Q8 — $g(x) = \ln(x^2 + 2x - 8)$, $h(x) = \ln(3x + 4)$, $g(x) < h(x)$

Domain: $x^2 + 2x - 8 = (x + 4)(x - 2) > 0$ gives $x < -4$ or $x > 2$; $3x + 4 > 0$ gives $x > -4/3$. So the domain is $x > 2$.

Drop the log: $x^2 + 2x - 8 < 3x + 4 \rightarrow x^2 - x - 12 < 0 \rightarrow (x - 4)(x + 3) < 0 \rightarrow -3 < x < 4$.

Intersect: with $x > 2$.

Answer: (2, 4)

Q9 — $p(x) = \log(x^2 + 3x + 1) - \log(2x + 5)$, $p(x) > 0$

Domain: $2x + 5 > 0$ gives $x > -5/2$. $x^2 + 3x + 1 > 0$ gives $x < (-3 - \sqrt{5})/2 \approx -2.62$ or $x > (-3 + \sqrt{5})/2 \approx -0.38$. Together: $x > (-3 + \sqrt{5})/2$. (The lower piece is empty — -2.62 is below $-5/2$.)

Combine the logs: $\log((x^2 + 3x + 1) / (2x + 5)) > 0$, so $(x^2 + 3x + 1) / (2x + 5) > 1$.

Solve: on the domain $2x + 5 > 0$, multiply through: $x^2 + 3x + 1 > 2x + 5 \rightarrow x^2 + x - 4 > 0 \rightarrow$ roots $(-1 \pm \sqrt{17})/2$, so $x < (-1 - \sqrt{17})/2$ or $x > (-1 + \sqrt{17})/2$.

Intersect: with the domain $x > (-3 + \sqrt{5})/2$.

Answer: $((-1 + \sqrt{17})/2, \infty)$

Q10 — $f(x) = \ln(x^2 - 5x + 4)$, $g(x) = \ln(2x - 2)$, $f(x) > g(x)$

Domain: $x^2 - 5x + 4 = (x - 1)(x - 4) > 0$ gives $x < 1$ or $x > 4$; $2x - 2 > 0$ gives $x > 1$. So the domain is $x > 4$.

Drop the log: $x^2 - 5x + 4 > 2x - 2 \rightarrow x^2 - 7x + 6 > 0 \rightarrow (x - 1)(x - 6) > 0 \rightarrow x < 1$ or $x > 6$.

Intersect: with $x > 4$.

Answer: (6, ∞)

All answers match the verified key in log-inequalities-nov3/answer-key.md (each checked by dense numeric sampling with the domain enforced).